

Vibe Coding – Lab 4 Chat History

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Project: 63_anvil_dodge

Assigned Repository: SETAPESU26/63_anvil_dodge

Personal Submission Repository: https://github.com/yashwanth00000/LAB1_AM818_SE

Purpose: Record of the Vibe Coding prompts and workflow used to fix the broken game and implement the README tasks.

1. Repository and Initial Setup

The assigned repository was inspected and identified as a Pygame Anvil Dodge game. The required workflow was to clone/fork the repository, run the Python game, record the original behavior for 10 seconds, then use Vibe Coding prompts to fix the bug and implement the listed features one by one.

2. Task 1 – Fix Player Boundary Bug

Vibe Coding prompt used:

Read the existing code carefully before making changes.

Fix Task 1 from the README: the player can move outside the visible left and right edges of the game window.

Requirements:

- Clamp the player's horizontal position to the visible screen boundaries.
- The entire player must remain visible.
- Preserve the existing Left/A and Right/D controls.
- Do not change unrelated gameplay.
- Do not implement any other README task yet.

After making the change, explain which file/function you modified and why.

Commit: Fix player off-screen boundary bug

3. Task 2 – Dynamic Difficulty Scaling

Vibe Coding prompt used:

Now implement Task 2 from the README: dynamic difficulty scaling.

First inspect the existing game architecture and understand how anvils are spawned and how their falling speed is controlled.

Implement progressive difficulty based on survival time.

Requirements:

- The game should begin at its current reasonable difficulty.
- As survival time increases, the game should progressively become harder.
- Increase anvil falling speed and/or spawning frequency gradually.
- Difficulty should increase smoothly rather than suddenly.
- Do not make the game impossible too quickly.
- Keep the player boundary fix unchanged.
- Keep collision detection, Game Over, timer and restart functionality working.
- Do not implement Task 3 or Task 4 yet.
- Make minimal, clean changes to the existing architecture.

After implementation, explain exactly how difficulty scales with survival time.

Commit: Implement dynamic difficulty scaling

4. Task 3 – Speed-Based Anvil Warning Tints

Vibe Coding prompt used:

Now implement Task 3 from the README: speed-based anvil warning tints.

Inspect the existing Anvil class and rendering code.

Requirements:

- Slow falling anvils should retain their normal appearance.
- Medium-speed anvils should receive a warning tint.
- High-speed anvils should receive a stronger hot warning color.
- The color must be determined from the anvil's current falling velocity.
- Use clear velocity thresholds.
- The visual change should make faster anvils appear more dangerous.
- Do not modify collision behavior.
- Do not modify player movement.
- Preserve Tasks 1 and 2.
- Do not implement Task 4 yet.

After implementation, explain the velocity thresholds and how the tinting works.

Commit: Add speed-based anvil warning tints

5. Task 4 – Ground Impact FX and Screen Shake

Vibe Coding prompt prepared for the final task:

Now implement the final task from the README: ground impact FX and screen shake.

Inspect the existing game loop and anvil ground-collision/removal logic.

Requirements:

- Detect when an anvil reaches the ground.
- Create a localized dust/particle effect at the impact position.
- Add a subtle short-duration screen shake.
- The effect should trigger when an anvil impacts the ground.
- Particles should disappear naturally after a short duration.
- Screen shake must be temporary and must not permanently move the game view.
- Do not interfere with player collision detection.
- Keep the game responsive and performant.
- Preserve all previously implemented features:
 1. Player boundary fix
 2. Dynamic difficulty scaling
 3. Speed-based warning tints
 4. Existing Game Over and restart behavior
- Do not rewrite unrelated code.

After implementing the changes, explain exactly what was modified and how the impact effect works.

6. Git Commit History

The first three tasks were committed separately as required by the lab:

Fix player off-screen boundary bug
Implement dynamic difficulty scaling
Add speed-based anvil warning tints

The fourth task should also be committed separately after implementation:

```
git add .  
git commit -m "Add ground impact FX and screen shake"  
git push origin main
```